

CLIENT DELIVERABLE

SAOS Service Manual v6.0

Systack Documentation

ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b
Success	Green	 #22c55e
Error	Red	 #ef4444
Accent highlights	Purple	 #8b5cf6

SAOS Service Manual

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What’s New in v2.1 (June 29, 2026)

Platform Updates

UPDATE	IMPACT	STATUS
Integration Health Monitoring	Real-time status of 5 core services	 Live
Universal Search	Search across tasks, messages, activity	 Live

UPDATE	IMPACT	STATUS
Usage Metrics Dashboard	Monthly stats + tier limit tracking	✓ Live
Trust Features	6 transparency sections	✓ Live
Security Hardening	Rate limiting, CORS, audit logging	✓ Live
Data Export	ZIP download of all client data	✓ Live
PIN Management	Full onboarding + change + reset flow	✓ Live

Service Tiers

Business Tier — \$299/month

Infrastructure: - Cloud VPS (16GB RAM) - Local Ollama (qwen2.5:7b) - Tailscale encrypted tunnel - n8n (10,000 runs/month) - Fully managed by Systack - Up to 5 team members

Agent Fleet (7 agents):

AGENT	ROLE	CAPABILITIES
SOL	Strategic oversight	High-leverage decisions, optimization
ASSEMBLY	Architecture	System design, builds, scaffolding
CHATTY	Communications	External comms, customer-facing
ATLAS	Research	Deep research, competitive analysis
GENI	Creative	Images, design, frontend
JURIS	Legal/Compliance	Legal review, compliance checks
PESSI	Monitoring	Alert triage, health reports

Services (8): 1. Invoice Processing Pipeline 2. Lead Qualification System 3. Customer Support Drafting 4. Document Classification Engine 5. Scheduled Report Generator 6. 7-Agent AI Fleet 7. n8n Workflow Hosting 8. Tailscale VPN + PostgreSQL

Support: - Email + Slack - Same business day response - 99.5% uptime SLA

Enterprise Tier — \$799/month

Infrastructure: - Cloud VPS (32GB RAM) - Local Ollama (llama3:70b) - Tailscale encrypted tunnel - n8n (unlimited runs) - Fully managed by Systack - Unlimited team members - Dedicated support line, 4hr SLA

Agent Fleet (10 agents): All Business agents PLUS: - CODY — Code review and validation - VALI — Testing and QA

Additional Services: - Unlimited n8n Workflows - 32GB RAM Infrastructure - Dedicated Support Line - Quarterly Business Reviews

Private Tier — \$799/month

Infrastructure: - Client-owned hardware (Mac Studio / Linux + RTX 4090) - Air-gapped or firewalled LAN - ZERO cloud apps (Stripe only, client account) - Self-hosted n8n

Services (6): 1. Private Document Extraction Pipeline 2. Automated Invoice Processing System 3. Self-Hosted Customer Support Automations 4. Local Data Entry Elimination System 5. Private Knowledge Base Search 6. Automated Compliance Audit Trail

Accelerate Tier — \$249/month

Infrastructure: - Cloud GPU instances (RunPod/Lambda) - Auto-scaling GPU - Tailscale WireGuard tunnel - n8n (10,000 runs/month)

Services (8): 1. Automated Invoice Processing System 2. Self-Hosted Customer Support Automations 3. Local Data Entry Elimination System 4. Automated Lead Qualification Pipeline 5. Document Classification & Routing Engine 6. Scheduled Report Generator 7. n8n Workflow Hosting 8. Tailscale VPN + PostgreSQL

Integration Health Monitoring

The dashboard now monitors 5 core integrations in real-time:

INTEGRATION	CHECK METHOD	HEALTHY WHEN
PostgreSQL	SELECT 1 query	Response < 1000ms
n8n	HTTP GET /healthz	HTTP 200
Tailscale	<code>tailscale status</code> subprocess	Exit code 0
BlueBubbles	HTTP GET to server root	HTTP 200
Ollama	HTTP GET /api/tags	HTTP 200

Status Levels: - ● Healthy — Responding within expected time - ● Degraded — Responding but slow or with errors - ● Down — Not responding or timeout

Overall Health: - healthy — All integrations healthy - degraded — At least one degraded - down — At least one down

Security Architecture (v2.1)

Authentication

- **PIN-based** — 4-10 digit PIN per client
- **Bearer tokens** — 32-byte URL-safe, 30-day expiry
- **Token storage** — SHA-256 hash in `client_auth_tokens` table

Rate Limiting

- 5 login attempts per 5 minutes per IP + client ID

- Exceeding returns HTTP 429 with attempt count
- Rate limit window resets after 5 minutes

CORS Policy

Authorized origins only: - `http://localhost:8768` - `https://*.ts.net` - `https://systack.net`

Audit Logging

All security events logged to `audit_log` table: - Login success/failure (with reason) - PIN changes - Token revocation - Data exports - Rate limit triggers

Token Revocation

Automatically revoked on: - PIN change (all tokens) - Logout (current token) - Token expiry (30 days)

Data Export

Clients can export all their data at any time:

Endpoint: `POST /api/export/data` (requires auth)

ZIP Contents:

```
saos-export-<client>-<date>.zip
├─ tasks/
│   └─ tasks.json          - All task history
├─ chat/
│   └─ conversations.json - All conversations + messages
├─ deliverables/
│   └─ (all files)         - Uploaded deliverables
├─ settings/
│   └─ client.json         - Account configuration
└─ README.txt              - Format guide
```

Privacy: Data is client-owned. No vendor lock-in.

Universal Search

Endpoint: `GET /api/portal/search?q=<query>` (requires auth)

Search Scope: - Tasks — `display_name`, `description`, `task_type` (ILIKE) - Chat Messages — `content` (ILIKE) - Activity Log — `action`, `entity_type` (ILIKE)

Response:

```
{
  "query": "invoice",
  "results": {
    "tasks": [...],
    "messages": [...],
    "activity": [...]
  },
}
```

```
"total": 5
}
```

Frontend: Debounced search (300ms, min 3 chars) with grouped dropdown results.

Service Agent Assignment

SERVICE	ASSIGNED AGENT	TASK TYPE
Invoice Processing	ASSEMBLY	service_setup
Lead Qualification	ASSEMBLY	service_setup
Customer Support	CHATTY	service_setup
Document Classification	ASSEMBLY	service_setup
Report Generation	ASSEMBLY	service_setup
Data Entry Elimination	ASSEMBLY	service_setup
Knowledge Base Search	ATLAS	service_setup
Document Extraction	ASSEMBLY	service_setup
Compliance Audit Trail	JURIS	service_setup
Infrastructure Setup	ASSEMBLY	service_setup
Dedicated Support	PESSI	service_setup

n8n Integration

Email Dispatcher Workflow

- Polls for pending email notifications every 60 seconds
- Sends via SMTP (credentials configured)
- Status: Active (may need port fix: 8768 → 8765)

Task Webhook

- `POST /api/internal/update-task/<id>` — Agents update task status
- `POST /api/chat/webhook/agent-response` — Agents send chat replies
- `GET /api/internal/pending-tasks` — Poll for pending tasks (requires internal API key)

Internal API Key

Set via environment variable:

```
SA0S_INTERNAL_API_KEY=<your-key>
```

Default (dev only): `saos-internal-dev-key`

Notification System

Channels

CHANNEL	WHEN USED
Dashboard Chat	Default — in-app messages
Email	Normal priority notifications
iMessage (BlueBubbles)	Urgent/critical notifications

Notification Flow

1. Task status change → System message in chat
2. If client offline → Queue notification
3. Urgent (FAILED status) → iMessage via BlueBubbles
4. Normal → Email via n8n SMTP workflow

Troubleshooting

Common Issues

ISSUE	CAUSE	FIX
Can't log in	Wrong PIN or Client ID	Verify credentials, use Forgot PIN
Rate limited	5+ failed attempts in 5 min	Wait 5 minutes
Integration shows down	Service not running	Check service, refresh health
Search no results	Query < 3 chars	Type at least 3 characters
Data export empty	No tasks/chat for client	Normal for new accounts
PDF won't load	File missing	Contact support

BlueBubbles Connection

- Server: `http://phillips-macbook-air.tail573d57.ts.net:1234`
- Phone: `+15012746231`
- If down: Check `brew services list | grep bluebubbles`

n8n Connection

- Server: `http://localhost:5678`
- Health: `GET /healthz`
- If down: Check n8n service status

API Endpoints Reference

Public (No Auth)

METHOD	ENDPOINT
GET	/api/portal/health
POST	/api/auth/login
POST	/api/auth/register
POST	/api/auth/forgot-pin
GET	/api/portal/onboarding-status/
GET	/api/portal/agents

Authenticated (Bearer Token)

METHOD	ENDPOINT
GET	/api/auth/me
POST	/api/auth/logout
POST	/api/auth/change-pin
GET	/api/portal/status
GET	/api/portal/tasks
GET	/api/portal/services
GET	/api/portal/activity
GET	/api/portal/deliverables
GET	/api/portal/integrations
GET	/api/portal/search?q=
GET	/api/chat/conversations
POST	/api/chat/conversations
GET	/api/chat/conversations//messages
POST	/api/chat/conversations//messages
POST	/api/chat/conversations//close
GET	/api/chat/poll
POST	/api/tasks/request
POST	/api/export/data
GET	/download/
GET	/api/deliverables/download/

Internal (X-Internal-API-Key)

METHOD	ENDPOINT
GET	/api/internal/pending-tasks
POST	/api/internal/claim-task/
POST	/api/internal/update-task/
POST	/api/chat/webhook/agent-response
POST	/api/internal/notify-client
GET	/api/internal/notifications/pending
POST	/api/internal/deliverables/upload